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Indirect Reciprocity Enhances Collective Cooperation on Weighted Networks

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***Abstract*—Direct, indirect, and network reciprocities are established mechanisms that sustain cooperation in natural and artificial systems. Yet which mechanism most effectively promotes cooperation on a given network remains unclear. Here, we develop a game-theoretic model to explore the evolution of direct and indirect reciprocity on weighted networks. Unlike classical donor–recipient frameworks, we study symmetric repeated interactions on undirected weighted networks with bilateral reputation updates, capturing heterogeneous tie strengths and accelerating reputation spread. We derive a general condition for reciprocal cooperation that unifies unweighted and weighted cases. Across large ensembles of random and empirical networks, indirect reciprocity consistently enhances cooperation, whereas stronger interactions sharply lower the benefit-to-cost threshold under direct reciprocity. To test the robustness of these insights, we examine competition among six reciprocity strategies and find that indirect reciprocity dominates. Our findings demonstrate that choosing the right reciprocity can promote global cooperation on social networks.**

***Index Terms*—Complex networks, cooperation, evolutionary game theory, reciprocity.**

I. Introduction

COOPERATION is an essential organizing principle vital to the prosperity of human societies, where individuals voluntarily incur private costs to benefit the collective [1], [2], [3], [4]. Although cooperation offers social advantages, natural selection inherently opposes it in the absence of an established mechanism for the evolution of cooperation [5]. Three prominent mechanisms are direct, indirect, and network reciprocity [6]. These reciprocal mechanisms, while interconnected in nature, exhibit significant differences in promoting cooperation [7], [8], [9]. Understanding the internal mechanisms for reciprocity to evolve remains an ongoing challenge [10].

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Direct reciprocity means that an individual's behavior toward partners depends on their recent interactions [7], [11], [12]. This mechanism necessitates a pair of individuals to interact repeatedly, fostering high levels of cooperation [7]. From evolutionary game theory, the prisoner's dilemma (PD) game is an effective tool for studying the evolution of cooperation [13]. In a one-shot PD game, natural selection favors defection [5]. Nevertheless, this preference shifts in repeated games [14], where individuals adjust their decisions based on the past behavior of their coplayers. Decision-making here relies on conditional cooperation strategies, such as *tit-for-tat* [15], *grim trigger* [16], and *win-stay lose-shift* [17] used to stimulate cooperation. Direct reciprocity is effective but fragile: when repeated interaction breaks, cooperation may quickly collapse [18].

Indirect reciprocity fosters cooperation through reputational information derived from third-party interactions, rather than direct personal experience [11], [19], [20]. Unlike direct reciprocity, it does not require repeated encounters between the same individuals [19]. Helping raises one's reputation, which in turn increases the chance of being helped by others [21]. Reputation is typically assigned based on simple social norms that classify behavior as good or bad [22]. Common strategies include *image scoring* [23], *shunning* [24], and *stern judging* [20].

A recent work has extended this framework to more realistic settings. Kawakatsu et al. [25] developed a mechanistic model of gossip-based reputation exchange, revealing how communication patterns and noise shape cooperative outcomes. Schnell and Muthukrishna [26] showed that misaligned reputations across nested social groups can destabilize cooperation at the global level. Wei et al. [27] introduced the notion of collective reputation in public goods games, demonstrating how group-level evaluation criteria influence the stability of cooperation.

Together, these findings highlight that indirect reciprocity is strongly influenced by the structure of communication, the scale of social interaction, and the strictness of normative rules. Yet most models assume homogeneous populations. How reputation-based cooperation operates in heterogeneous societies is less understood [28]. Network reciprocity offers a promising framework for exploring these dynamics.

Network reciprocity [13], [29], [30], [28], [31] signifies that individuals interact with only their neighbors, where neighboring ones interact more frequently than others due to network structure. As mentioned above, direct and indirect reciprocity rely on a common assumption of a well-mixed population,

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where everybody interacts equitably. Unfortunately, real-world population (or network) structures are not well-mixed [32]. Network reciprocity seeks to determine under what conditions, and which networks facilitate or inhibit cooperation [6]. Pioneering research highlights that cooperation thrives in unweighted regular networks when an individual's neighbors are rare [13]. For any weighted network, cooperation flourishes most when individuals have strong pairwise relationships [29]. Even conjoining segregated cooperation-inhibiting networks, if done correctly, can save global cooperation from collapse [31]. Therefore, it is self-evident that network structure significantly influences the emergence of cooperation. As pioneering works, all these results were derived by focusing on the competition between unconditional cooperators and defectors.

Recent research has explored the evolution of cooperation through direct, indirect, and network reciprocity. Direct reciprocity emphasizes the role of repeated interactions in fostering cooperation [7], [14]. Indirect reciprocity, driven by reputation-based norms, promotes cooperation without repeated interactions between the same individuals [19], [22]. Network reciprocity highlights the impact of network structure on cooperative behavior [29], [31]. However, most studies have focused on a single reciprocity mechanism or assumed well-mixed populations. They overlook how direct and indirect reciprocity evolve across different network structures, particularly in weighted networks.

This gap motivates a unified approach to reciprocity on weighted networks. We introduce a game-theoretic model that embeds both direct and indirect reciprocity in repeated, symmetric interactions on undirected, weighted graphs, where edge weights reflect heterogeneous interaction strengths. Unlike well-mixed or donor–recipient formulations, our framework uses bilateral reputation updates that incorporate both first-hand and third-party information, thereby quantifying how network structure and information flow jointly shape incentives for cooperation.

Our analysis extends prior work [33] on reciprocity in directed or asymmetric settings by providing a tractable, symmetric framework that yields explicit invasion conditions. Classical models of indirect reciprocity assume asymmetric roles for donors and recipients. In contrast, our setting features mutual action and bilateral reputation updates. This symmetry accelerates the spread of reputations across weighted ties and, as our results demonstrate, reduces the critical benefit-to-cost ratio required for cooperation, especially when strong ties are concentrated on a few edges.

The main contribution of this work is that we develop a game-theoretic model to encapsulate the evolution of direct and indirect reciprocity on weighted networks. Drawing on our model, we derive a critical benefit-to-cost threshold for the success of reciprocal cooperation. In addition, we provide mathematical results applicable to any payoff structure, birth–death updating, and mutation. A thorough analysis of large-scale random networks and six empirical social networks demonstrates that indirect reciprocity strongly enhances cooperation. Conversely, as interactions intensify, we find a dramatic decrease in the critical threshold for sustaining cooperation through direct reciprocity. Reciprocity, emerging across various societal domains, thus serves as a fertile ground for cooperation to flourish.

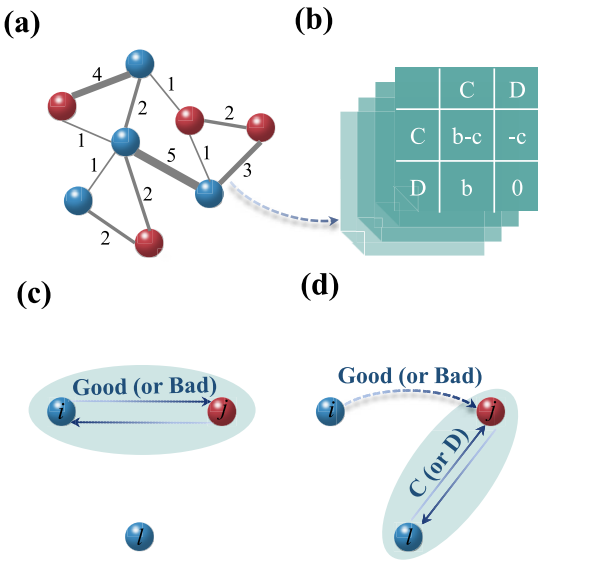

Fig. 1. Evolutionary dynamics of reciprocity on weighted networks. (a) Population structure is represented by an undirected, weighted network, where nodes correspond to individuals and edges represent interactions between them. The edge weights e_{ij} reflect the strength of these interactions. (b) We model a repeated PD game in which individuals decide whether to cooperate based on the behavior of the previous round. Cooperation incurs a cost c and provides a benefit b to the neighbor, while defection entails no cost or benefit. Individuals assign reputations to their neighbors, classifying them as good or bad. (c) In direct reciprocity ($\lambda = 0$), reputation assignments are based solely on personal experiences. (d) In indirect reciprocity ($\lambda = 1$), reputation assignments also consider third-party perspectives. Individuals typically cooperate with good neighbors and defect against bad ones, guided by direct interactions and observed reputations.

The remainder of this article is structured as follows. Sections II–IV outline the underlying principles of our model, focusing on reciprocity coevolution, reputation updating, and strategy evolution. Sections V–VII derive the simple condition for the emergence of reciprocal cooperation on any network. Section VIII presents simulation results regarding the evolution of direct and indirect reciprocity on weighted networks. Finally, Section IX concludes with a discussion of our findings.

II. Problem Formulation

We consider repeated PD games played on a population of size N (Fig. 1). The population structure is represented as an undirected, weighted network $G = (V_e, E_d, A_m)$, with edge weights e_{ij} [Fig. 1(a)]. The set of nodes is represented by $V_e = 1, 2, \ldots, N$, and the set of edges is $E_d \subseteq (i, j) : i, j \in V_e, j \neq i$. The adjacency (weight) matrix $A_m = [a_{ij}]$ satisfies $a_{ij} = e_{ij}$ if $(i, j) \in E_d$ and $a_{ij} = 0$ otherwise. A_m is symmetric, we have ${A_m}^T = A_m$. Each individual is represented as a node in the network, classified as either a cooperator (A) or a defector (B). The edges indicate that individuals interact with their k neighbors [Fig. 1(b)], where k denotes the degree of the network. The interactions are governed by the following universal payoff matrix:

$$\begin{array}{c} \\ A \\ B \end{array} \begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \begin{pmatrix} R & S \\ T & P \end{pmatrix} \end{array}. \tag{1}$$

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In this payoff matrix (1), the four elements represent the payoffs for different outcomes in the PD game. When both the individuals cooperate (top-left), they receive the reward R. If one cooperates while the other defects (top-right and bottom-left), the cooperator incurs a cost S, while the defector reaps the temptation payoff T. If both defect (bottom-right), both the individuals receive the punishment payoff P. We assume $T > R > P > S$ and $2R > T + S$, ensuring a strict PD. This reflects that cooperation is appealing, but defection yields a higher payoff in the short term [5].

For a specific variation in the PD game [29], the corresponding payoff matrix is

$$\begin{array}{c} \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cc} C & D \end{array} \\ \begin{pmatrix} b-c & -c \\ b & 0 \end{pmatrix} \end{array}. \tag{2}$$

In this version of the game (2), when both the individuals cooperate (top-left), the cooperator pays a cost c and provides a benefit b to the other. If one defects while the other cooperates (top-right and bottom-left), the cooperator suffers a penalty of $-c$, while the defector receives the benefit b. If both the individuals defect (bottom-right), neither gains any benefit, resulting in a payoff of 0. This payoff structure is commonly used to model situations where cooperation entails a cost but benefits others.

In each round, a pair of adjacent individuals is randomly selected to play one round of the PD game [Fig. 1(a) and (b)]. Their choices are binary: cooperate (C) or defect (D), made independently. Cooperation means paying a cost $c > 0$ to provide a benefit $b > c$ to the neighbor, while defection means paying no cost and providing no benefit. All the actions are transparent to the neighbors, but the perception error ε may influence their interpretation. Specifically, third-party observers may misinterpret a defector as a cooperator with probability ε, and vice versa. Each individual assigns a reputation (i.e., good or bad) based on their strategy and interactions with neighbors.

After each interaction, there is a probability d that a different pair of neighboring individuals will be selected for another round of the PD game. With probability $1 - d$, interactions stop. In addition, there is a probability d_c that the same two individuals will engage in a subsequent round. Once interactions cease, each individual receives an edge-weighted average payoff from all the rounds they participated in.

Our model extends previous work on reciprocity coevolution, exploring how cooperation emerges through direct, indirect, and network reciprocity [11], [20], [29]. These models focus on the emergence of collective behavior through individual interactions. In our setup, each individual's decision to cooperate or defect is influenced by their perception of neighbours' reputations. These reputations evolve dynamically based on the observed actions of others, with higher payoff strategies invading the population. Social learning and behavioral imitation further facilitate the spread of such strategies.

The model incorporates both direct and indirect reciprocity. Direct reciprocity occurs when individuals update their reputations based solely on direct experiences, while indirect reciprocity involves reputation updates based on third-party interactions. These two types of reciprocity are captured by the strategy space, represented by a four-tuple (u, y, z, λ). The parameter u defines the baseline probability of treating a neighbor as good in the absence of prior information, while y and z represent conditional probabilities for interacting with cooperators and defectors, respectively. Specifically, with probability y (or z), an individual who has cooperated (or defected) is considered good. The final parameter, λ, determines the reliance on indirect interactions for reputation updates. When $\lambda = 0$, individuals rely solely on their own experiences [direct reciprocity, Fig. 1(c)] [15], whereas for $\lambda = 1$, they incorporate third-party perspectives [indirect reciprocity, Fig. 1(d)] [23].

Indirect reciprocity plays a crucial role in larger populations, where direct interactions are often infrequent or entirely absent. In this context, individuals update their reputations based on the observed actions of third parties rather than direct interactions. This mechanism facilitates the emergence of cooperation in large populations, even when direct encounters are rare or absent. In the framework outlined by [23], individuals make decisions based on the reputation of others, which is shaped by third-party observations. The parameter λ in our model represents how individuals rely on indirect information. When $\lambda = 0$, decisions are based solely on direct experiences (direct reciprocity), whereas $\lambda = 1$ means individuals consider all available information, including third-party reputations (indirect reciprocity). This approach is particularly relevant in real-world scenarios where direct interactions are rare, but reputation-based systems, such as online reviews or feedback, guide decision-making. For example, in social media or online platforms, users decide whether to cooperate or trust others not only based on their own direct experiences but also on the reputation established through feedback from other users, illustrating indirect reciprocity. By leveraging the actions and reputations of others, indirect reciprocity promotes cooperation even in large, complex networks where direct interaction is minimal, allowing prosocial behavior to emerge through social learning and information diffusion.

This framework allows us to explore a wide range of strategies for cooperation, including well-known strategies such as *tit-for-tat* [15] TFT = $(1, 1, 0, 0)$, *win-stay-lose-shift* [17] WSLS = $(1, 0, 1, 0)$, *grim trigger* [16] GT = $(1, 0, 0, 0)$, *image scoring* [23] IS = $(1, 1, 0, 1)$, *stern judging* [20] SJ = $(1, 0, 1, 1)$, and *shunning* [24] SH = $(1, 0, 0, 1)$. In all these strategies, cooperating with a good neighbor results in a good reputation, whereas defecting against a good neighbor results in a bad reputation. Individuals are inclined to cooperate with good neighbors and defect against bad ones. By integrating direct and indirect reciprocity, our model offers a more comprehensive understanding of how cooperation can evolve on any network structure.

To more clearly illustrate the cooperative incentives of these strategies, we show a comparison of the evolutionary trajectories of cooperation for different strategies. Simulation results confirm the superiority of the *shunning* strategy, which can successfully dominate the population. In general, indirect reciprocity outperforms direct reciprocity in most network structures. For further details, see Section VIII.

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III. Updating Reputations

Following a round of pairwise game interactions, populations update their perspectives on everyone's reputations. Concretely, a random interaction between the first pair of individuals is sampled from the most recent round, and an individual's reputation is evaluated based on the action toward the neighbor. Explicitly, consider a population encompassing precisely two strategies, indicated as $s = 2$. Initially, an individual i chosen at random interacts with $\sum_{j\neq i} a_{ij}$ neighbors. For weighted networks, we set $a_{ij} = 1$ if $a_{ij} > 0$. As outlined previously, the probability of a specific individual being selected to participate in the next round of the PD game is $\bar{\theta} = \sum_{j\neq i} a_{ij}/N$, leading to $\theta = \sum_{j\neq i} a_{ij}/[N(N-1)]$ as the probability for a specific pair of individuals selection. We denote the set of adjacent A-players with strategy $A = (u_1, y_1, z_1, \lambda_1)$ as k_A, and the adjacent B-players with strategy $B = (u_2, y_2, z_2, \lambda_2)$ as $\sum_{j\neq i} a_{ij} - k_A$. The term $r_1 = y_1 - z_1$ is defined. Let $P_{AB}(t)$ represent the probability of A-player regarding B-player as good at time t. Each individual can obtain a good reputation in the following avenues P_{AA}, P_{AB}, P_{BA}, and P_{BB}, and the probability is thus given by

$$\begin{aligned}\mathbf{P}_t &= (P_{AA}, P_{AB}, P_{BA}, P_{BB})^T \\ &= (\mathbf{1} - d\cdot\mathbf{M})^{-1}\left[(1-d)\,\mathbf{P}_0 + d\boldsymbol{\eta}\right].\end{aligned} \tag{3}$$

Herein, the identity matrix is indicated as $\mathbf{1}$. Vectors $\mathbf{P}_0$ and $\boldsymbol{\eta}$ have dimensions s^2. We describe them as $\mathbf{P}_0 = (u_1, u_1, u_2, u_2)^T$ and $\boldsymbol{\eta} = (\eta_1, \eta_1, \eta_2, \eta_2)^T$, with entries

$$\begin{aligned}\eta_1 &= \theta z_1 + \lambda_1\left(\bar{\theta} - \theta\right)(\varepsilon r_1 + z_1) \\ \eta_2 &= \theta z_2 + \lambda_2\left(\bar{\theta} - \theta\right)(\varepsilon r_2 + z_2).\end{aligned} \tag{4}$$

$\mathbf{M}$, a $s^2 \times s^2$ matrix, is articulated as

$$\mathbf{M} = \begin{pmatrix} m_{11} & m_{12} & 0 & 0 \\ 0 & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & 1-\theta-\lambda_2\left(\bar{\theta}-\theta\right) & 0 \\ 0 & 0 & \theta\lambda_2 r_2\,(1-2\varepsilon)\,k_A & m_{44} \end{pmatrix} \tag{5}$$

where

$$\begin{aligned}m_{11} &= 1-\theta-\lambda_1\left(\bar{\theta}-\theta\right) + \theta r_1\left(1+\lambda_1\,(1-2\varepsilon)\,(k_A-2)\right) \\ m_{12} &= \theta\lambda_1 r_1\,(1-2\varepsilon)\left(\sum_{j\neq i} a_{ij} - k_A\right) \\ m_{22} &= 1-\theta-\lambda_1\left(\bar{\theta}-\theta\right) \\ m_{23} &= \theta r_1\left(1+\lambda_1\,(1-2\varepsilon)\,(k_A-1)\right) \\ m_{24} &= \theta\lambda_1 r_1\,(1-2\varepsilon)\left(\sum_{j\neq i} a_{ij} - k_A - 1\right) \\ m_{31} &= \theta\lambda_2 r_2\,(1-2\varepsilon)\,(k_A-1) \\ m_{32} &= \theta r_2\left(1+\lambda_2\,(1-2\varepsilon)\left(\sum_{j\neq i} a_{ij} - k_A - 1\right)\right) \\ m_{44} &= 1-\theta-\lambda_2\left(\bar{\theta}-\theta\right) + m_{32} - \theta r_2\lambda_2\,(1-2\varepsilon).\end{aligned} \tag{6}$$

Since θ and $\bar{\theta}$ depend on the network's degree $\sum_{j\neq i} a_{ij}$, P_{AB} and P_{BA} are related to not only the reciprocity strategy but also the network structure. The detailed derivation of (3)–(6) can be found in literature [11]. Even so, this analysis is divergent from the research on direct and indirect reciprocity in a well-mixed population [11], because reputations are constantly renovated in a weighted network.

IV. Payoff and Strategy Evolution

Each individual receives a benefit, b, from interactions with neighbors who perceive them positively. In addition, individuals incur a cost, c, only when interacting with good neighbors. The average edge-weighted payoff to individual i, considering all the pairwise interactions and evolving reputations, is given by

$$\pi_i = \frac{b\sum_{j\in G}\left(e_{ij}\big/\sum_l e_{il}\right)s_j\cdot\left(1-P_{ji}\right) - cs_i\cdot P_{ij}}{\sum_{j\neq i} a_{ij} - 1} \tag{7}$$

where s_i and s_j indicate the type of individual: 1 for A-individuals and 0 for B-individuals. If individuals i and j adopt strategies $A = (u_1, y_1, z_1, \lambda_1)$ and $B = (u_2, y_2, z_2, \lambda_2)$, then $P_{ij} = P_{AB}$ and $P_{ji} = P_{BA}$, as described in (3). This equation simplifies the computation of payoffs across the population.

In (7), P_{AB} (or P_{BA}) represents the probability that A-individual (B-individual) views B-individual (A-individual) as a good neighbor. These probabilities are crucial for incorporating the evolving reputations of individuals, which influence their payoffs. The first term aggregates benefits from neighbors, weighted by tie strengths. The second term subtracts costs when cooperating with those deemed good.

In our model, we explore the evolutionary propagation of strategies across a population, using the death–birth (DB) updating [13]. After everybody's reputations are assigned by all the population members, a random individual following strategy B is initially chosen to update its strategy, potentially adopting the strategy A of a neighbor in proportion to the payoff. Under DB updating, a random node is replaced and neighbors compete to fill the vacancy with probability proportional to $e_{ij}R_i$. The reproductive rate for individual i is $R_i = 1+\delta\pi_i$, where $\delta > 0$ measures the strength of selection. In scenarios of weak selection, $\delta \ll 1$. Therefore, if j is chosen to be replaced, the probability that i reproduces into this vacancy is

$$\mathbb{R}(i \to j) = e_{ij}R_i \Big/ \sum_{l\in G} e_{lj}R_l. \tag{8}$$

This updating rule models how strategies spread within a population. In this process, one individual is randomly selected for replacement, and its strategy may be updated based on the reproductive success of its neighbors. The reproductive rate, R_i, depends on individual i's payoff, π_i, and the strength of selection, δ. Higher payoffs, resulting from more successful strategies, increase the likelihood that an individual will reproduce and pass on her strategy. This mechanism mirrors natural selection, where individuals with advantageous strategies are more likely to influence the population. By incorporating this rule, we capture the evolution of strategies in a population, where reproductive success and network structure, through edge weights, are crucial in shaping the spread of cooperation or defection.

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V. Derivation of $(b/c)^*$

The coalescence time [29], [30] γ_{ij} is the expected meeting time of random walks from i and j, defined by the recurrence relations

$$\gamma_{ij} = 1 + \frac{1}{2}\sum_{l\in G, i\neq j}\left(e_{il}\gamma_{jl} + e_{jl}\gamma_{il}\right). \tag{9}$$

For weighted networks without self-loops, we have $\gamma_{ii} = 0$ and $e_{ii} = 0$ [29]. Pairwise statistics of the coalescence time can be obtained by

$$\left[\!\left[\frac{1}{N} - s_i s_j\right]\!\right] = \frac{\gamma_{ij}}{2N} \quad i, j \in G. \tag{10}$$

Here, $[\![]\!]$ denotes an expectation over a particular state distribution under neutral drift, $\delta = 0$.

Let γ_i denote the expected coalescence time for two independent random walks from i to rejoin each other. It is obtained by the relation

$$\gamma_i = 1 + \sum_{j\in G} q_{ij}\gamma_{ij} \tag{11}$$

where $q_{ij} = e_{ij}/\sum_l e_{il}$. We define the expected coalescence time from the two ends of an n-step random walk as $\gamma^{(n)} = 1/N\sum_{i,j\in G} {q_{ij}}^{(n)}\gamma_{ij}$, satisfying the recurrence relation

$$\gamma^{(n+1)} = \gamma^{(n)} + \frac{1}{N}\sum_{i\in G} {q_{ij}}^{(n)}\gamma_i - 1. \tag{12}$$

For weighted regular networks, we observe that $q_{ij}^{(0)} = 1$, $q_{ij}^{(1)} = 0$ and $q_{ij}^{(2)} = \sum_{i\in G} q_{ij}^2$ is the Simpson degree of G [40]. From (12), we have

$$\begin{aligned}\gamma^{(1)} &= N - 1\\ \gamma^{(2)} &= N - 2\\ \gamma^{(3)} &= N + N\sum_{i\in G} q_{ij}^2 - 3.\end{aligned} \tag{13}$$

The fixation probability, ψ_A (or ψ_B), connotes the probability of a stochastic A-player (or B-player) generating a lineage that invades the entire population. Applying recent results on evolutionary dynamics [29], [30], we revise the fixation probabilities in the game [see (2)] as

$$\begin{aligned}\psi_C &= \frac{1}{N} + \frac{\delta}{2N}\left[-c\gamma^{(2)}P_{CD} + b\left(\gamma^{(3)} - \gamma^{(1)}\right)\left(1 - P_{DC}\right)\right]\\ &\quad + O\left(\delta^2\right)\\ \psi_D &= \frac{1}{N} + \frac{\delta}{2N}\left[c\gamma^{(2)}P_{CD} - b\left(\gamma^{(3)} - \gamma^{(1)}\right)\left(1 - P_{DC}\right)\right]\\ &\quad + O\left(\delta^2\right).\end{aligned} \tag{14}$$

Based on (3), P_{CD} denotes the probability of cooperators regarding defectors as good, while P_{DC} represents the probability of defectors perceiving cooperators as good. For a sufficiently huge population and weak selection, cooperators are preferred to defectors if and only if $\psi_C > 1/N > \psi_D$ [13]. Therefore, substituting (14) directly derives (15) and (16) in Section VI.

VI. Condition for Facilitating Cooperation

From seminal research, Ohtsuki et al. [13] discovered that $b/c > k$ is a rudimentary rule for facilitating cooperation on unweighted networks. After this, Allen et al. [29] extend this rule to weighted networks. As a foundational step, they scrutinized the most basic interaction between unconditional cooperators and defectors, namely, ALLC and ALLD strategies. Yet, exploring which strategies of direct or indirect reciprocity emerge on any network structure remains captivating.

Our principal finding holds for any payoff matrix [see (1)], but we first study a PD game [see (2)]. Cooperators bear a cost, c, to supply a benefit, b. Defectors, on the other hand, neither bear a cost nor confer any benefit ($b > c > 0$). Given the absence of mutation and migration in our baseline model, we leverage the fixation probability to elucidate the impact of selection on a population [41]. Here, we focus on the competition between reciprocal cooperators (C) and always defectors (D), quantifying the fixation probabilities of ψ_C and ψ_D. To elucidate, ψ_C denotes the probability that a single cooperator adopting the reciprocity strategy disseminates its strategy across a monolithic population of defectors adopting ALLD strategy. Conversely, ψ_D signifies the probability that a single defector successfully invades $N - 1$ cooperative residents. For a significantly expansive population coupled with weak selection, the condition of reciprocal cooperators being preferred to defectors ($\psi_C > \psi_D$) is

$$-c\gamma^{(2)}P_{CD} + b\left(\gamma^{(3)} - \gamma^{(1)}\right)\left(1 - P_{DC}\right) > 0. \tag{15}$$

Intuitively, the aforementioned condition underscores the imperative that a cooperator must exhibit an elevated reproduction probability compared with their defective neighbors. This condition stems from characterizing the average payoff of an individual two steps away. The primary term, $-c\gamma^{(2)}P_{CD}$, intimates that an individual pays a cost c whenever it is a cooperator (expected time $\gamma^{(2)}P_{CD}$). The remaining term captures a benefit b from one-step cooperative neighbors [expected time $(\gamma^{(3)} - \gamma^{(1)})(1 - P_{DC})$]. Hence, reciprocal cooperation (that is, cooperation via direct or indirect reciprocity) is favored when the benefit-to-cost ratio breaches a threshold (and $\gamma^{(3)} > \gamma^{(1)}$)

$$\left(\frac{b}{c}\right)^* = \frac{\gamma^{(2)}P_{CD}}{\left(\gamma^{(3)} - \gamma^{(1)}\right)\left(1 - P_{DC}\right)}. \tag{16}$$

Herein, defection thrives for $\gamma^{(3)} < \gamma^{(1)}$ [30], [31]. Especially, (16) is a threshold condition for reciprocal cooperation in the iterated PD game to be favored by natural selection. The parameters, P_{CD} and P_{DC}, capture a strategy that leads to everyone adaptively assigning and modifying everyone else's reputation. Based on (3), $P_{CD}, P_{DC} \in [0, 1]$, which depend on the reciprocity strategy (e.g., TFT and IS), denote the probability of cooperators (or defectors) perceiving defectors (or cooperators) as good. Each individual tends to cooperate with good opponents, yet defect with bad opponents in every round of the iterated PD game. In this case, a cooperator adopts the strategy $A = (u_1, y_1, z_1, \lambda_1)$, while defectors adopt the strategy $B = (u_2, y_2, z_2, \lambda_2)$. $\gamma^{(n)}$ is the expected coalescence time from the two ends of an n-step random walk (see Methods),

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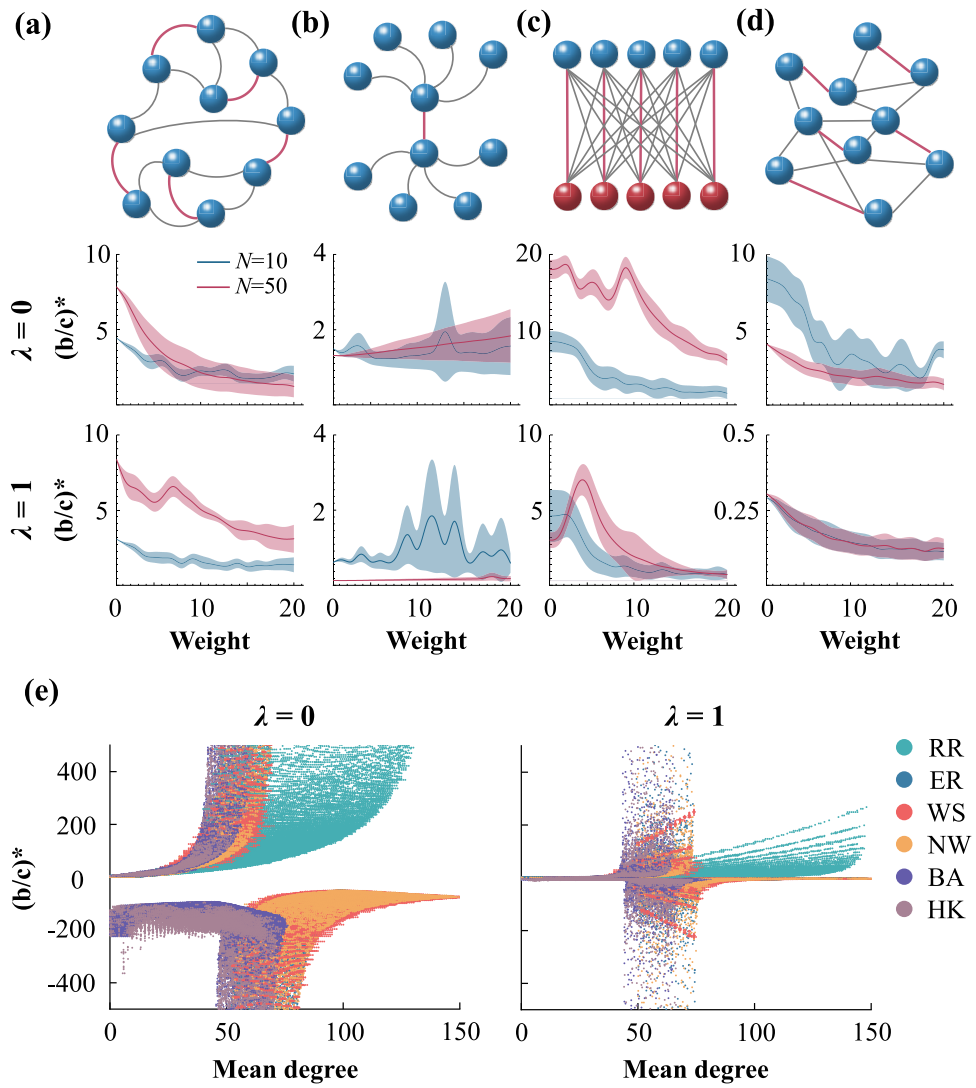

Fig. 2. Networks that promote cooperation through direct and indirect reciprocity. Indirect reciprocity is more conducive to cooperation thriving on any network structure. (a)–(d) Each individual has one neighbor with weight e_{ij}. All others have weight 1. Herein, any cooperative behavior is favored for substantial weight. This is exemplified through various examples such as a weighted regular network (a), a weighted star-joining network (b), a weighted bipartite network (c), and a weighted BA network (d). Each point represents an average over 10^2 Monte Carlo simulations. (e) Conditions for reciprocal cooperation on 0.6 million random networks. Direct reciprocity produces more negative $(b/c)^*$ values for more realistic networks. While positive $(b/c)^*$ values of indirect reciprocity are significantly lower than direct reciprocity, especially for regular networks. The critical ratio $(b/c)^*$ is computed by solving (16) (no Monte Carlo simulations). Initially, a cooperator is randomly positioned to invade a population of $N-1$ defectors. Direct reciprocity corresponds to TFT = $(1,1,0,0)$, and IS = $(1,1,0,1)$ for indirect reciprocity. Initial network sizes are uniformly sampled in the range $100 \le N \le 300$. Random network models' parameters are as follows: Erdös–Rényi (ER) [34] with edge probability $0 < p_e < 1$; random regular (RR) [35] with node degree $1 \le k \le 3N/4$; Watts–Strogatz (WS) [36] with nearest neighbors $1 \le k \le 3N/4$ and edge rewiring probability $0 < p_e < 0.05$; Newman–Watts (NW) [37] is consistent with that of WS; Barabási–Albert (BA) [38] with linking number $1 \le m \le 10$; and Holme–Kim (HK) [39] with linking number $2 \le m \le 4$ and triad formation parameter $0 < p_t < 0.15$. We use $d_c = 0.5$, $\varepsilon = 0$, and $\delta = 0.01$.

relying upon the network structure. Surprisingly, by assuming that cooperators adopt the strategy ALLC $= (1,1,1,\lambda)$ and defectors adopt the strategy ALLD $= (0,0,0,\lambda)$, (16) is streamlined to the condition $b/c > t_2/(t_3 - t_1)$ on weighted networks [29]. In unweighted k-regular networks, (16) reduces to the recognized principle $b/c > k$ [13]. Subsequently, we will verify the proposed condition via simulations on an assortment of random networks.

VII. Long-Term Stability and Sustainability of Cooperation

In this section, we extend the analysis of cooperation from Section VI to examine its long-term stability and sustainability. While Section VI outlines the conditions necessary for cooperation to evolve, our focus is on whether cooperative strategies can endure over time within a networked population, influenced by direct and indirect reciprocity.

The evolutionary dynamics of cooperation depend on the fixation probabilities of cooperators and defectors in the population. As previously derived, the fixation probability of a cooperator, ψ_C, represents the likelihood that a single cooperator will spread and eventually dominate the entire population. Conversely, the fixation probability of a defector, ψ_D, is the probability that defectors will take over the population. Cooperation will be sustained long-term if $\psi_C > \psi_D$.

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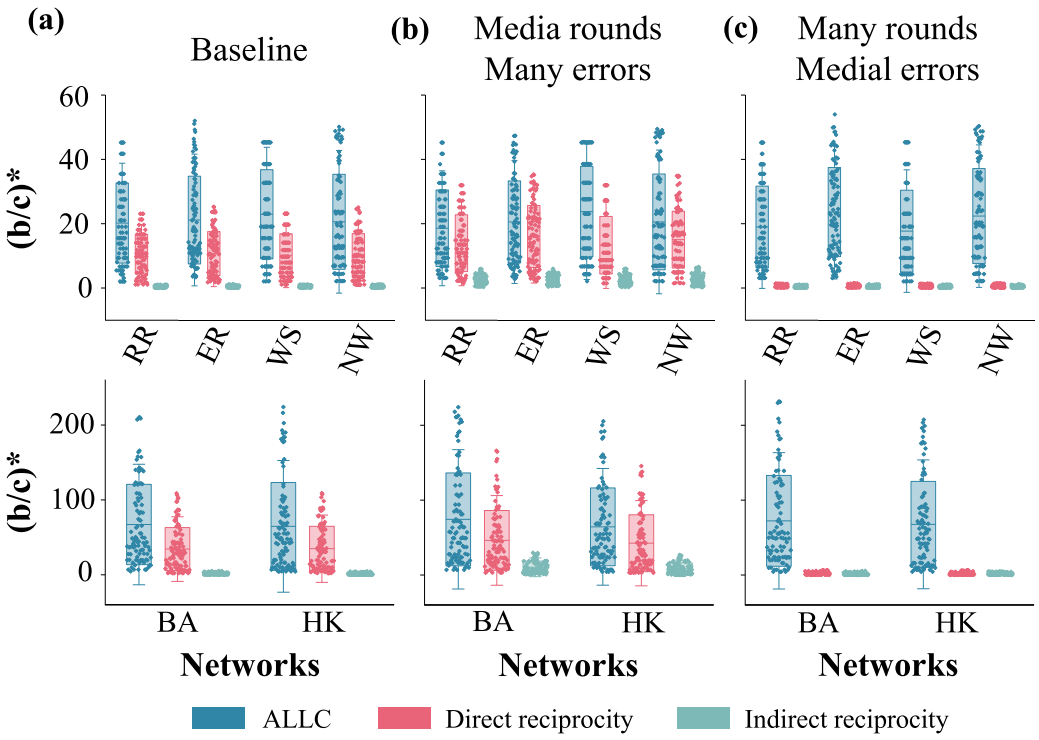

Fig. 3. Effects of environmental noise on cooperation. (a)–(c) For sufficient interactions ($d_c \to 1$), there is a dramatic decrease in the critical $(b/c)_0^*$ for direct reciprocity ($\lambda = 0$). Amplifying the perception error subtly elevates the threshold of $(b/c)_1^*$ for indirect reciprocity ($\lambda = 1$). Using (16), we compare the critical $(b/c)^*$ of ALLC, direct and indirect reciprocity strategies on the above networks. Each box plot represents a random sample of 10^2 from 10^4 Monte Carlo simulations. The scenarios differ in the game's continuation probability d_c and the perception error ε. The specific parameters we use are $d_c = 0.5$, $\varepsilon = 0$ (a), $d_c = 0.9$, $\varepsilon = 0.1$ (b) and $d_c = 0.99$, and $\varepsilon = 0.01$ (c). The rest of the parameters are consistent with Fig. 2.

To identify the conditions for this outcome, we examine the reproductive advantages of cooperators relative to defectors. For large populations and weak selection pressures, the fixation probability of a cooperator can be approximated by: $\psi_C = 1/N + \delta/(2N)[-c\gamma^{(2)}P_{CD} + b(\gamma^{(3)} - \gamma^{(1)})(1 - P_{DC})] + O(\delta^2)$. The $\gamma^{(n)}$ represents the coalescence time for an n-step random walk on the network. The terms P_{CD} and P_{DC} denote the probabilities that a cooperator views a defector as beneficial and vice versa, reflecting the impact of reciprocity mechanisms. Cooperation will persist if the critical benefit-to-cost ratio satisfies: $b/c > (\gamma^{(2)}P_{CD})/()(\gamma^{(3)} - \gamma^{(1)})(1 - P_{DC})$. This criterion provides insight into the long-term sustainability of cooperation: it remains stable if the reproductive advantage of cooperators outweighs that of defectors, which is influenced by the network structure, reciprocity strategy, and the benefit-to-cost ratio.

The network structure directly affects the coalescence times $\gamma^{(2)}$ and $\gamma^{(3)}$, which are critical for determining fixation probabilities. In highly connected networks, the interaction distance between individuals is shorter, making it more likely that cooperators will interact with one another, thus reinforcing cooperative behaviors. In contrast, sparse networks with low connectivity increase the expected coalescence times, making cooperation harder to sustain unless the benefit-to-cost ratio is sufficiently high.

To further illustrate this, consider the expression for the effective coalescence time $\gamma^{(n)}$ for two random walkers on the network, which depends on the network topology: $\gamma^{(n+1)} = \gamma^{(n)} + 1/N \sum_{i \in G} q_{ij}{}^{(n)}\gamma_i - 1$. $q_{ij}^{(n)}$ is the transition probability for an n-step random walk from node i to node j, and γ_i is the expected coalescence time for node i. Networks with higher connectivity tend to have smaller values for $\gamma^{(2)}$ and $\gamma^{(3)}$, thus reducing the required benefit-to-cost ratio for cooperation to remain stable.

VIII. Results

Here, we present the results of our simulations and experiments, in which we systematically examine the impact of various parameters such as edge weights, network topologies, average degree, and perception errors in promoting cooperation. The chosen experimental parameters encompass a broad range of real-world network structures and cooperation scenarios, as outlined in the literature [13], [29]. Edge weights e_{ij} were selected to represent different interaction strengths between individuals, while the average degree k_m enabled us to explore sparsely and densely connected networks. Furthermore, we varied the perception error ε and pairwise continuation probability d_c to investigate the effects of noisy environments and the information reliability on the evolution of cooperation.

A. Random Network Investigations

In Fig. 2(a)–(d), we present the evolution of cooperation in weighted regular networks, star networks, bipartite networks, and BA networks. Our data show that in regular networks,

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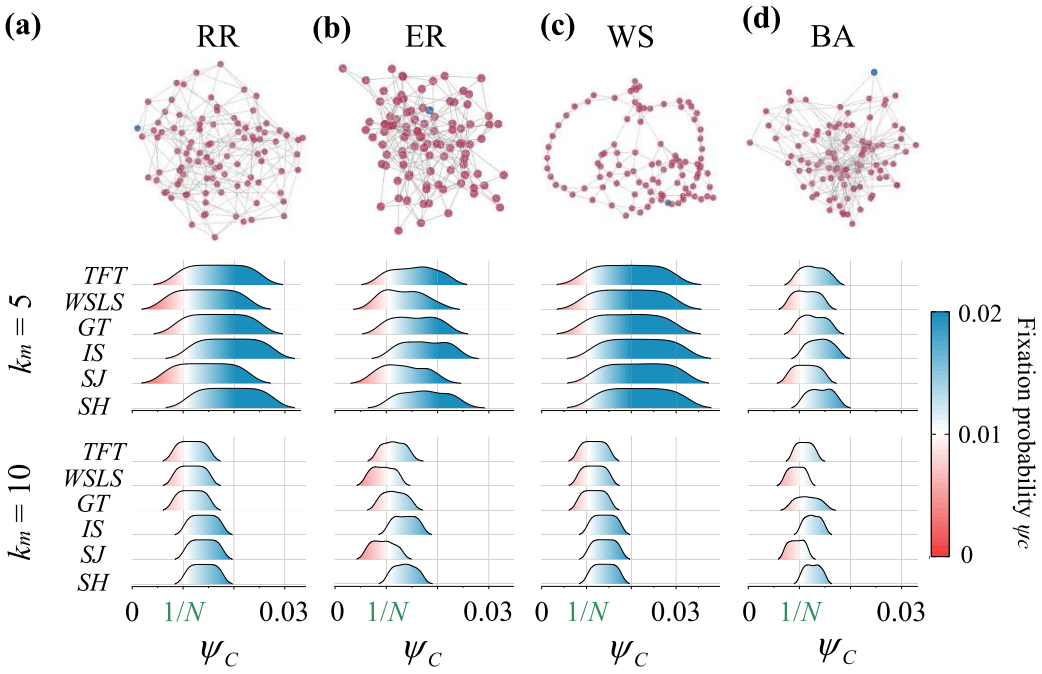

Fig. 4. Fixation probabilities of competing reciprocity strategies. (a)–(d) *Shunning* is the most powerful strategy against other strategies, having the highest region of $\psi_C > 1/N$ against any other strategy. Generally, *shunning* and *image scoring* always outperform other strategies, whereas in regular networks (a) and (c), *stern judging* approaches the performance of the first two strategies. Here, we compute the fixation probabilities of six reciprocity strategies for promoting cooperation on (a) RR, (b) ER, (c) WS, and (d) BA networks. The fixation probability is computed using (14). If $\psi_A > 1/N > \psi_B$, natural selection favors strategy A. For the sake of comparison, ψ_C represents the fixation probability that a single cooperator adopting one of such reciprocity strategies invades $N-1$ defectors adopting strategy ALLD. Each subplot represents the kernel smooth distribution of $\psi_C \in [0, 0.02]$ over 50 Monte Carlo simulations. The second and third rows, respectively, display numerical simulations for mean degree $k_m = 5$ and $k_m = 10$. Generally, we find that the fixed probabilities of IS and SH are greater than $1/N = 0.01$, which means that the red area corresponding to $0 < \psi_C < 0.01$ disappears. Examples of those strategies include TFT = $(1, 1, 0, 0)$, WSLS = $(1, 0, 1, 0)$, GT = $(1, 0, 0, 0)$, IS = $(1, 1, 0, 1)$, SJ = $(1, 0, 1, 1)$, and SH = $(1, 0, 0, 1)$. We use $N = 100$ and $1 \le b/c \le 20$. Other parameters are consistent with Fig. 2.

node degrees are relatively uniform. However, sudden changes in edge weights can still trigger local cooperation. In contrast, star and BA networks display marked topological heterogeneity due to the presence of highly connected hub nodes. Specifically, in the heterogeneous BA networks, these hubs serve as key conduits for information and reputation. This facilitates the rapid spread of indirect reciprocity over short distances and lowers the critical $(b/c)^*$ for cooperation. Moreover, the high connectivity of central nodes in star networks further enhances reputation transmission. The data shown in each subplot, averaged over 10^2 Monte Carlo simulations, clearly demonstrate how cooperation flourishes on networks with substantial edge weights.

As shown in Fig. 2(e), we computed $(b/c)^*$ for 600 000 random networks, each drawn from six distinct network models. The results robustly support the dominance of indirect reciprocity, revealing how cooperation becomes increasingly difficult as the average degree rises. We observe that across different random network models, the critical $(b/c)^*$ for direct reciprocity shifts toward more negative values as the average degree increases. This suggests that in highly connected networks, local reciprocity tends to evolve into punitive behavior due to increased intergroup competition. In contrast, the critical $(b/c)^*$ for indirect reciprocity remains relatively low. In small-world and scale-free networks, high clustering and long-range connections enable efficient reputation transmission. Even with heterogeneous degree distributions, indirect reciprocity can lower the cost-to-benefit ratio needed for cooperation. These findings indicate that topological features play a decisive role in the effectiveness of direct and indirect reciprocity strategies.

B. Environmental Noise Effect

We investigate how environmental noise affects the evolution of cooperation by analyzing the critical $(b/c)^*$ values for three strategies: ALLC, direct reciprocity (implemented via *tit-for-tat*), and indirect reciprocity (implemented via *image scoring*). Fig. 3(a)–(c) shows our systematic investigation of how the continuation probability d_c and perception error ε influence cooperation. Our data, obtained from 10^4 Monte Carlo simulations and 10^2 random samples per condition, reveal that noise more severely disrupts cooperation in regular networks than in scale-free networks. The latter demonstrates a higher resilience to environmental disturbances. In our analysis, indirect reciprocity consistently outperforms the other strategies even in noisy settings. However, higher perception errors challenge cooperative behavior. Networks with high connectivity and scale-free properties sustain cooperation better despite increased ε.

Fig. 3 illustrates that when d_c is low, individuals tend to defect because the incentive to cooperate is weak without the prospect of future reciprocation. As

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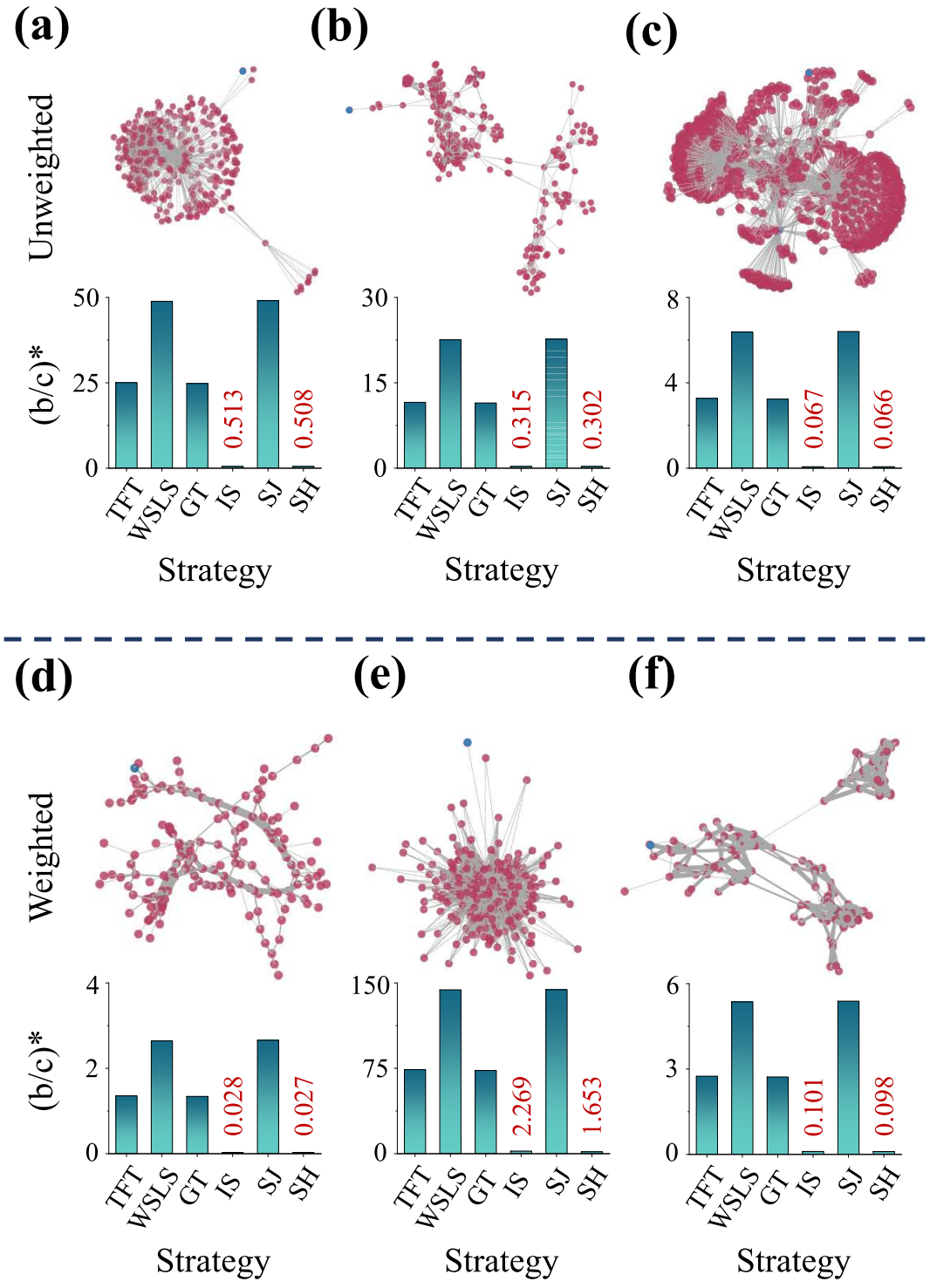

Fig. 5. Conditions for reciprocal cooperation on six empirical social networks. (a)–(f) *Shunning* always has a lowest $(b/c)^*$ against other strategies, which once again proves the power of *shunning* from an empirical view. Here, we compute the critical $(b/c)^*$ for such strategies on six empirical social networks. The critical $(b/c)^*$ is solved by (16), which means that the threshold condition for cooperators adopting one of such reciprocity strategies to be favored over defectors adopting strategy ALLD. Examples of those networks include (a) social ego network from Facebook, with nodes $N = 4039$ and edges $E_d = 88\,234$; (b) malaria var DBLa HVR network, with $N = 307$ and $E_d = 2812$; (c) word adjacency network in texts of Japanese, with $N = 3177$ and $E_d = 8300$; (d) India bus network in Mumbai, with $N = 2266$, $E_d = 3042$, and weights $1 \le e_{ij} \le 35$; (e) London's bicycle sharing network, with $N = 750$, $E_d = 1\,469\,851$ and $1 \le e_{ij} \le 3197$; and (f) friendship relations network in a Marseilles' high school, with $N = 327$, $E_d = 1808$, and $1 \le e_{ij} \le 10$. Other parameters are consistent with Fig. 4.

d_c approaches 1, cooperation stabilizes, and the critical $(b/c)^*$ values drop for both direct and indirect reciprocity. Notably, indirect reciprocity remains more effective at higher d_c values, even under significant noise.

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The decision rules underlying *tit-for-tat* and *image scoring* yield distinct responses to environmental noise. The *tit-for-tat* strategy, which relies solely on the opponent's most recent action, is susceptible to even minor perception errors. Such errors may be misinterpreted as defection, triggering a cascade of retaliatory defections that sharply increase the critical value $(b/c)^*$. In contrast, the *image scoring* strategy builds a cumulative reputation over multiple interactions. This mechanism reduces the impact of isolated misperceptions and maintains a lower, more stable $(b/c)^*$ even as noise increases. Thus, as shown in Fig. 3, *image scoring* sustains cooperation under high continuation probabilities d_c and elevated perception errors ε, whereas *tit-for-tat* quickly loses its cooperative stability.

Our extensive Monte Carlo simulations capture the differential impact of environmental noise on *tit-for-tat* and *image scoring*. These findings explain the results in Fig. 3 and suggest that the robustness of indirect reciprocity in noisy environments stems from its ability to buffer against sporadic errors. Future work should extend this stability analysis to a broader range of noise levels and more complex settings.

C. Competing Reciprocity Strategies

We examine the fixation probabilities of six different reciprocity strategies across various network types (Fig. 4). Each subfigure presents data from 50 Monte Carlo simulations, illustrating how each strategy performs across different network topologies. We find that *shunning* (SH) consistently outperforms all other strategies, exhibiting the highest fixation probability $\psi_C > 1/N$. This result holds across different benefit-to-cost ratios b/c and average degrees k_m, underscoring the robustness of indirect reciprocity.

Despite *shunning*'s dominance, it is crucial to explore other strategies to gain a fuller understanding of reciprocity mechanisms. Specifically, *image scoring* (IS) is particularly effective in heterogeneous networks, such as scale-free networks, where individuals rely heavily on reputation to guide cooperation. This strategy benefits from long-range connections, enabling individuals to draw on indirect observations from their neighbors. In contrast, *stern judging* (SJ) performs better in regular networks, where the network structure promotes equal opportunities for all the individuals. Its stricter judgment ability allows it to perform similar to *shunning* in some cases.

Finally, we compute the critical $(b/c)^*$ for various reciprocity strategies across six real-world social networks (Fig. 5). These networks include examples such as Facebook, Mumbai's bus network, and London's bicycle-sharing system. The results from these real-world networks further support the superiority of *shunning* in promoting cooperation. We find that *shunning* consistently has the lowest $(b/c)^*$ value, confirming its power across diverse environments. However, our analysis also highlights the potential of *image scoring* and *stern judging* to foster cooperation in specific network types, suggesting that they should not be overlooked in studies of reciprocity dynamics. This work analyzed empirical social network datasets from the University of Colorado Boulder.

D. Sensitivity Analysis of Key Parameters

To evaluate the robustness of our findings, we conducted a sensitivity analysis to examine how key parameters such as network topology, edge weights, average degree, and perception errors affect the evolution of cooperation. Specifically, we assessed the impact of these parameters on the critical benefit-to-cost ratio $(b/c)^*$, which determines whether cooperation or defection prevails in different network structures.

For instance, in Fig. 2(e), we calculated $(b/c)^*$ for 600 000 random networks drawn from six distinct network models, including Erdős-Rényi (ER), WS, and BA. These models represent a range of topological properties. Our analysis showed that as the average degree of the network increases, the critical $(b/c)^*$ value becomes more difficult for cooperation to thrive, particularly in networks based on direct reciprocity. In networks with small-world and scale-free properties, direct reciprocity [represented by the strategy TFT $= (1, 1, 0, 0)$] yields more negative $(b/c)^*$ values, indicating that cooperation is less likely as network connectivity increases.

In contrast, indirect reciprocity [represented by the strategy IS $= (1, 1, 0, 1)$] consistently supports cooperation, even in complex networks with high average degrees. This is further demonstrated in Fig. 3, where we show that indirect reciprocity remains effective even as the perception error ε increases. As ε rises, the critical $(b/c)^*$ for cooperation shifts, but indirect reciprocity continues to outperform direct reciprocity, especially when the continuation probability d_c is high. Our analysis also confirms that networks with higher edge weights and greater average degrees, such as scale-free and small-world networks, are more conducive to cooperation. In Fig. 2(a)–(d), we observe that networks with significant edge weights (e_{ij}) enhance cooperation by making interactions more rewarding, particularly when individuals are more likely to cooperate with neighbors who have higher edge weights. This is reflected in the fact that the critical $(b/c)^*$ for indirect reciprocity is lower compared with direct reciprocity in these settings.

These results demonstrate that indirect reciprocity remains robust across various network topologies and outperforms direct reciprocity, especially in networks where complexity and noise make cooperation more difficult to sustain.

IX. Discussion and Conclusion

Despite focusing on the PD game, our model can be extended to any payoff structure (see (1) in Methods). Our general result (16) is not confined to any particular updating rule or strategic interaction and can be applied to birth–death updating and mutation. For each of these variations, we derive the conditions for reciprocal cooperation in terms of coalescence times and reputation assessment. This means that the competing strategies of direct or indirect reciprocity can instantiate repeated interactions and conditional behavior on social networks.

Based on six representative strategies (**TFT, WSLS, GT, IS, SJ,** and **SH**), our simulations indicate that *shunning* consistently promotes cooperation. These strategies collectively represent a concise yet comprehensive spectrum of reciprocity rules. Although our results show *shunning* outperforming alternative norms across various networks, a nuanced interpretation

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is necessary given differing informational assumptions in prior studies.

Specifically, *shunning* and *stern judging* prove effective in our model due to symmetric, undirected networks where reputation updates rely on globally accessible but potentially noisy observations. This setup resembles semi-public assessments rather than strictly private or group-based ones. In contrast, previous research indicates that strict norms diminish under private observations. Okada et al. [42] analytically demonstrated that neither *shunning* nor *stern judging* achieves cooperative evolutionary stability when assessments occur privately. Similarly, Pandula et al. [43] found that *shunning* is less effective in environments involving perception errors and abductive reasoning.

Our framework addresses these limitations by assuming partial reputation sharing and symmetric interaction rules, enabling the reliable propagation of strict norms. Furthermore, although Kessinger et al. [20] reported *stern judging*'s success in group-structured populations, their outcome depended on intergroup norm competition and migration, which our model does not incorporate. Such discrepancies underscore that the evolutionary success of social norms hinges on context: strict norms thrive in structured, transparent settings, whereas tolerant norms may prevail in fragmented or noisy environments. Thus, our findings complement existing literature by clarifying the specific conditions under which *shunning* and *stern judging* emerge as viable strategies.

Although our model effectively explores the evolution of reciprocal cooperation on networks, its applicability to larger networks remains challenging. The Monte Carlo simulation algorithm used in our study significantly increases computational costs as the network size grows. The time complexity of the simulation is $O(N \cdot t)$, where N represents the number of individuals in the network and t denotes the number of time steps. While the model is well-suited for smaller networks (e.g., $N \sim 10^3$), it becomes computationally demanding when applied to large-scale networks, such as those found in social media or communication systems. To enable the model's application to large-scale networks, future work may consider optimization strategies, such as parallelizing the Monte Carlo simulations. Using high-performance computing (HPC) clusters or cloud computing could distribute the computational load and reduce overall simulation time. In addition, approximation techniques such as mean-field approximations or variational Monte Carlo methods could minimize the number of required simulations, efficiently estimating the critical benefit-to-cost ratio b/c. To further enhance the model's applicability to large networks, we also plan to extend the model to account for dynamic network evolution and integrate optimization strategies with model accuracy. This will allow us to analyze large-scale networks with millions of nodes, as encountered in real-world scenarios.

Despite these advancements, our model has certain limitations. We assumed a static network with interactions solely based on reputation to reduce complexity and focus on the core finding that indirect reciprocity enhances network cooperation. However, real-world networks evolve, and this assumption may constrain the applicability of our results. Future work should incorporate dynamic network structures, account for real-world constraints such as limited information and decision biases, and integrate adaptive mechanisms (e.g., multiagent reinforcement learning and context-dependent reputation updates). Moreover, our current model does not consider mixed or adaptive strategies. We also plan to integrate this mechanism to capture the complexity of cooperative dynamics in evolving systems.

We acknowledge that our current model uses undirected and symmetrically weighted networks to ensure analytical tractability. Incorporating directed graphs and asymmetric weightings would more accurately reflect real-world social dynamics, especially concerning influential individuals. Preliminary analyses indicate that influential nodes could substantially impact cooperative outcomes by efficiently disseminating cooperative or punitive strategies. Directionality or asymmetry could thus significantly alter the critical benefit-to-cost threshold required for sustaining cooperation. Future research will systematically investigate these effects to fully quantify their implications.

In summary, we have derived a condition for determining how natural selection favors two competing reciprocity strategies on any network structure. Although our work is theoretically stimulated, it has essential implications for empirical research. Our result helps elucidate which reciprocity mechanisms best promote prosocial behaviors, such as cooperation. We demonstrate that indirect reciprocity outperforms direct reciprocity in most network structures, even with large perception errors.

REFERENCES

[1] A. Liu, L. Wang, G. Chen, and X. Guan, "Heterogeneously networked evolutionary games with intergroup conflicts," *IEEE Trans. Cybern.*, vol. 54, no. 10, pp. 5684–5695, Oct. 2024.

[2] W. Li et al., "Multiagent consensus tracking control over asynchronous cooperation–competition networks," *IEEE Trans. Cybern.*, vol. 55, no. 9, pp. 4347–4360, Sep. 2025.

[3] J. Shi et al., "Payoff control in multichannel games: Influencing opponent learning evolution," *IEEE Trans. Cybern.*, vol. 55, no. 2, pp. 776–785, Feb. 2025.

[4] W. Guan, X. Song, T. Gan, J. Lin, X. Chang, and L. Nie, "Cooperation learning from multiple social networks: Consistent and complementary perspectives," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4501–4514, Sep. 2021.

[5] R. Rider, "The evolution of cooperation," *Science*, vol. 5, nos. 3–4, pp. 406–409, Mar. 1984.

[6] M. A. Nowak, "Five rules for the evolution of cooperation," *Science*, vol. 314, no. 5805, pp. 1560–1563, Dec. 2006.

[7] J. Tkadlec, C. Hilbe, and M. A. Nowak, "Mutation enhances cooperation in direct reciprocity," *Proc. Nat. Acad. Sci. USA*, vol. 120, no. 20, May 2023, Art. no. e2221080120.

[8] Y. Fujimoto and H. Ohtsuki, "Evolutionary stability of cooperation in indirect reciprocity under noisy and private assessment," *Proc. Nat. Acad. Sci. USA*, vol. 120, no. 20, May 2023, Art. no. e2300544120.

[9] J. Li, X. Zhao, B. Li, C. S. L. Rossetti, C. Hilbe, and H. Xia, "Evolution of cooperation through cumulative reciprocity," *Nature Comput. Sci.*, vol. 2, no. 10, pp. 677–686, Oct. 2022.

[10] J. Gross, Z. Z. Méder, C. K. W. De Dreu, A. Romano, W. E. Molenmaker, and L. C. Hoenig, "The evolution of universal cooperation," *Sci. Adv.*, vol. 9, no. 7, Feb. 2023, Art. no. eadd8289.

[11] L. Schmid, K. Chatterjee, C. Hilbe, and M. A. Nowak, "A unified framework of direct and indirect reciprocity," *Nature Human Behaviour*, vol. 5, no. 10, pp. 1292–1302, May 2021.

[12] M. van Veelen, J. García, D. G. Rand, and M. A. Nowak, "Direct reciprocity in structured populations," *Proc. Nat. Acad. Sci. USA*, vol. 109, no. 25, pp. 9929–9934, Jun. 2012.

[13] H. Ohtsuki, C. Hauert, E. Lieberman, and M. A. Nowak, "A simple rule for the evolution of cooperation on graphs and social networks," *Nature*, vol. 441, no. 7092, pp. 502–505, May 2006.

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This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

[14] A. McAvoy, J. Kates-Harbeck, K. Chatterjee, and C. Hilbe, "Evolutionary instability of selfish learning in repeated games," *PNAS Nexus*, vol. 1, no. 4, Sep. 2022, Art. no. pgac141.
[15] M. A. Nowak and K. Sigmund, "Tit for tat in heterogeneous populations," *Nature*, vol. 355, no. 6357, pp. 250–253, Jan. 1992.
[16] D. Wu, M. Antony, and K. Y. Szeto, "Evolution of grim trigger in prisoner dilemma game with partial imitation," in *Applications of Evolutionary Computation EvoApplications* (Lecture Notes in Computer Science), vol. 6024, C. D. Chio et al., Eds., Berlin, Germany: SpringerIstanbul, Turkey. Springer, 2010, pp. 151–160.
[17] M. Nowak and K. Sigmund, "A strategy of win-stay, lose-shift that outperforms tit-for-tat in the Prisoner's dilemma game," *Nature*, vol. 364, no. 6432, pp. 56–58, Jul. 1993.
[18] C. Hilbe, L. A. Martinez-Vaquero, K. Chatterjee, and M. A. Nowak, "Memory-*n* strategies of direct reciprocity," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 18, pp. 4715–4720, Apr. 2017.
[19] C. Hilbe, L. Schmid, J. Tkadlec, K. Chatterjee, and M. A. Nowak, "Indirect reciprocity with private, noisy, and incomplete information," *Proc. Nat. Acad. Sci. USA*, vol. 115, no. 48, pp. 12241–12246, Nov. 2018.
[20] T. A. Kessinger, C. E. Tarnita, and J. B. Plotkin, "Evolution of norms for judging social behavior," *Proc. Nat. Acad. Sci. USA*, vol. 120, no. 24, Jun. 2023, Art. no. e2219480120.
[21] S. Pal and C. Hilbe, "Reputation effects drive the joint evolution of cooperation and social rewarding," *Nature Commun.*, vol. 13, no. 1, Oct. 2022, Art. no. 5928.
[22] F. P. Santos, F. C. Santos, and J. M. Pacheco, "Social norm complexity and past reputations in the evolution of cooperation," *Nature*, vol. 555, no. 7695, pp. 242–245, Mar. 2018.
[23] M. A. Nowak and K. Sigmund, "Evolution of indirect reciprocity by image scoring," *Nature*, vol. 393, no. 6685, pp. 573–577, Jun. 1998.
[24] A. L. Radzvilavicius, T. A. Kessinger, and J. B. Plotkin, "Adherence to public institutions that foster cooperation," *Nature Commun.*, vol. 12, no. 1, Jun. 2021, Art. no. 3567.
[25] M. Kawakatsu, T. A. Kessinger, and J. B. Plotkin, "A mechanistic model of gossip, reputations, and cooperation," *Proc. Nat. Acad. Sci. USA*, vol. 121, no. 20, May 2024, Art. no. e2400689121.
[26] E. Schnell and M. Muthukrishna, "Indirect reciprocity undermines indirect reciprocity destabilizing large-scale cooperation," *Proc. Nat. Acad. Sci. USA*, vol. 121, no. 19, Apr. 2024, Art. no. e2322072121.
[27] M. Wei et al., "Indirect reciprocity in the public goods game with collective reputations," *J. Roy. Soc. Interface*, vol. 22, no. 225, Apr. 2025, Art. no. 20240827.
[28] A. McAvoy, B. Allen, and M. A. Nowak, "Social goods dilemmas in heterogeneous societies," *Nature Human Behaviour*, vol. 4, no. 8, pp. 819–831, May 2020.
[29] B. Allen et al., "Evolutionary dynamics on any population structure," *Nature*, vol. 544, no. 7649, pp. 227–230, Mar. 2017.
[30] B. Allen, G. Lippner, and M. A. Nowak, "Evolutionary games on isothermal graphs," *Nature Commun.*, vol. 10, no. 1, Nov. 2019, Art. no. 5107.
[31] B. Fotouhi, N. Momeni, B. Allen, and M. A. Nowak, "Conjoining uncooperative societies facilitates evolution of cooperation," *Nature Hum. Behav.*, vol. 2, no. 7, pp. 492–499, Jul. 2018.
[32] Q. Su, A. McAvoy, Y. Mori, and J. B. Plotkin, "Evolution of prosocial behaviours in multilayer populations," *Nature Human Behav.*, vol. 6, no. 3, pp. 338–348, Jan. 2022.
[33] H. Wei, J. Zhang, and C. Zhang, "Evolutionary dynamics of direct and indirect reciprocities with directional interactions," *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 3, pp. 3243–3254, Jun. 2024.
[34] P. Erdős and A. Rényi, "On the evolution of random graphs," *Trans. Amer. Math. Soc.*, pp. 38–82, May 2011.
[35] N. C. Wormald, "Models of random regular graphs," *Surveys in Combinatorics* (London Mathematical Society Lecture Note Series), J. D. Lamb and D. A. Preece, Eds., Cambridge, U.K.: Cambridge Univ. Press, Oct. 1999, pp. 239–298.
[36] D. J. Watts and S. H. Strogatz, "Collective dynamics of 'small-world' networks," *Nature*, vol. 393, no. 6684, pp. 440–442, Jun. 1998.
[37] M. E. J. Newman and D. J. Watts, "Renormalization group analysis of the small-world network model," *Phys. Lett. A*, vol. 263, nos. 4–6, pp. 341–346, Dec. 1999.
[38] A.-L. Barabási and R. Albert, "Emergence of scaling in random networks," *Science*, vol. 286, no. 5439, pp. 509–512, Oct. 1999.
[39] P. Holme and B. J. Kim, "Growing scale-free networks with tunable clustering," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 65, no. 2, Jan. 2002, Art. no. 026107.
[40] B. Allen, J. Gore, and M. A. Nowak, "Spatial dilemmas of diffusible public goods," *eLife*, vol. 2, Dec. 2013, Art. no. e01169.
[41] N. Sharma and A. Traulsen, "Suppressors of fixation can increase average fitness beyond amplifiers of selection," *Proc. Nat. Acad. Sci. USA*, vol. 119, no. 37, Sep. 2022, Art. no. e2205424119.
[42] I. Okada, T. Sasaki, and Y. Nakai, "A solution for private assessment in indirect reciprocity using solitary observation," *J. Theor. Biol.*, vol. 455, pp. 7–15, Oct. 2018.
[43] N. Pandula, E. Akçay, and B. Morsky, "Indirect reciprocity with abductive reasoning," *J. Theor. Biol.*, vol. 580, Mar. 2024, Art. no. 111715.

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